



**INTERNATIONAL CAMPUS
OF SCIENCE AND TECHNOLOGY**

International Campus of Science and Technology

Faculty of Computing, Department of IT
HDIT Batch 01

HDIT - DBMS

Project Name: Restaurant Management System - Cafulon

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REQUIREMENT SPECIFICATION DOCUMENT

Cafulon Restaurant Management System

Module	HDIT 12093 — Database Management Systems
Group	Group Cafulon
Project	Design and Development of a Database Management System for a Real-World Organization
Organization	Cafulon Restaurant & Dining

1. Organization Background

Cafulon Restaurant & Dining is a mid-sized full-service restaurant located in Colombo, Sri Lanka, with a seating capacity of 60 guests across indoor and outdoor sections. Established in 2019, Cafulon serves a fusion of Sri Lankan and international cuisine, offering dine-in, takeaway, and delivery services. The restaurant operates with a staff of 24 employees including chefs, waiters, hostesses, and management personnel.

The restaurant currently manages its operations through manual record-keeping — paper-based order slips, handwritten reservation books, and spreadsheet-based accounting. As the business has grown, this manual approach has led to increasing operational inefficiencies, data inconsistencies, and customer service delays.

2. Problem Statement

Cafulon Restaurant faces the following operational challenges:

1. Order Management: Handwritten order slips are frequently lost or misread, leading to incorrect orders and customer dissatisfaction.
2. Reservation Tracking: The paper-based reservation book causes double-booking of tables and difficulty in tracking customer preferences.
3. Menu Management: Menu price updates require reprinting of physical menus; there is no centralized system to track item availability.
4. Payment Recording: Payment records are scattered across multiple notebooks, making revenue reconciliation time-consuming and error-prone.
5. Reporting: Generating daily sales reports, identifying top-selling items, and tracking employee performance requires manual data aggregation spanning hours of work.
6. Data Redundancy: Customer and employee information is duplicated across multiple records, leading to inconsistency.
7. No Centralized Database: There is no single source of truth for restaurant data, making it difficult to retrieve historical information or generate insights.

A database management system is required to centralize all restaurant data, eliminate redundancy, enforce data integrity, and provide real-time operational visibility.

3. Project Objectives

1. To analyze and document the business requirements of Cafulon Restaurant's daily operations.
2. To design a normalized Entity-Relationship (ER) diagram capturing all entities, attributes, and relationships.
3. To apply normalization techniques up to 3NF to eliminate data redundancy.
4. To implement the database using MySQL with proper constraints (PK, FK, CHECK, UNIQUE, NOT NULL).
5. To create a minimum of 20 sample records per main table for realistic testing.
6. To develop 10 SQL queries covering CRUD operations, joins, grouping, and aggregate functions.
7. To build a web-based front-end application for data visualization and management.
8. To test and validate the system through structured test cases.

4. Scope of System

The Cafulon Restaurant Management System covers the following functional areas:

Area	Details
Customer Management	Registration, contact details, VIP status tracking
Employee Management	Staff records, positions, salary, shift assignment
Menu Management	Items, categories, pricing, availability, preparation time
Table Management	Table numbers, capacity, location, active/inactive status
Reservation Management	Booking, date/time, party size, special requests, status tracking
Order Processing	Order creation with multiple items, order type, status workflow
Payment Processing	Amount, method, transaction reference, payment status
Reporting	Daily sales, top-selling items, employee performance, revenue analysis

Out of Scope

- Payroll processing
- Inventory/supplier management
- Online ordering portal
- Mobile application

5. Functional Requirements (Min. 10)

ID	Requirement	Description
FR-01	Add Customer Records	System shall allow registration of new customers with name, phone, email, address, and city
FR-02	Update Customer Records	System shall allow editing of existing customer information
FR-03	Delete Customer Records	System shall allow deletion of customer records with referential integrity checks
FR-04	Add Employee Records	System shall record employee details including position, salary, hire date, and shift
FR-05	Manage Menu Items	System shall allow adding, updating, and removing menu items with pricing and availability
FR-06	Create Reservations	System shall allow booking tables with customer, date, time, party size, and status tracking
FR-07	Process Orders with Multiple Items	System shall allow creating orders with multiple menu items, auto-calculating subtotals and total
FR-08	Track Order Status	System shall support order status workflow: Pending → Preparing → Served → Completed / Cancelled
FR-09	Record Payments	System shall record payments against completed orders with method and transaction reference

FR-1 0	Generate Daily Sales Report	System shall provide aggregate reports showing daily revenue, order count, and average order value
FR-1 1	View Active Menu Items	System shall display all available menu items with their current prices
FR-1 2	Search Reservations by Date	System shall allow filtering reservations by date for daily planning

6. Non-Functional Requirements (Min. 5)

ID	Requirement	Description
NFR-0 1	Performance	System shall respond to queries within 2 seconds for datasets up to 10,000 records
NFR-0 2	Data Integrity	System shall enforce referential integrity through foreign key constraints
NFR-0 3	Availability	System shall be available during restaurant operating hours (10:00–23:00) on the local network
NFR-0 4	Security	System shall use parameterized queries to prevent SQL injection; database user shall have least-privilege access
NFR-0 5	Usability	Web interface shall be accessible on desktop and tablet devices with minimal training required

NFR-06	Scalability	Database schema shall support future addition of new menu items, tables, and employees without structural changes
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7. User Types

User Type	Description	Access Rights
Administrator/Manager	Restaurant owner or general manager	Full access: add/edit/delete all records, view reports, manage employees
Staff (Waiters/Hostess)	Front-of-house staff handling orders and reservations	Create and update orders and reservations; view customer and menu data
Kitchen Staff	Chefs and kitchen helpers	View incoming orders and update order status to Preparing/Served
Cashier	Staff handling bill payments	Record payments, view completed orders, generate receipts